HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology
Semester Examination 2025

Course Code: MAT323
Midterm Examination
Date: 6/21/2025

Total Time: 1Hour 30 Minutes

Course Title: Numerical Analysis
Semester: Spring 2025
Total Marks: 30

Answer any three (03) out of the following questions: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.			CLOs	Marks
Q1.	(a)	Define interpolation	CLO1	[1]
	(b)	Derive the Newton's Gregory formula for forward interpolation.	CLO2	[5]
	(c)	The Gamma function is tabulated for various values of x in the following table:	CLO3	[4]

x	1.15	1.16	1.17	1.18	1.19
$\Gamma(x)$	0.93304	0.92980	0.92670	0.92373	0.92088

Interpolate the values of $\Gamma(1.1859)$.

Q2.	(a)	Let $y = f(x)$ denotes a function with takes the values $y_0, y_1, \cdots y_n$ for the equidistant values $x_0, x_1, \cdots x_n$ of the independent variable x, then what is the value of $\Delta^2 y_0$?	CLO1	[1]
	(b)	Construct a divided difference table for the following values:	CLO3	[2]

x	1	2	4	7	12
$f(x)$	22	30	82	106	216

	(c)	Derive the Newton's General interpolation formula for unequal intervals.	CLO2	[4]
	(d)	Show that $\Delta^3_{bcd}\left(\frac{1}{a}\right) = -\frac{1}{abcd}$	CLO3	[3]
Q3.	(a)	Define algebraic and transcendental equation with examples.	CLO1	[2]
	(b)	Derive the iterative method to solve the equation $f(x) = 0$.	CLO2	[4]
	(c)	Find the root of the equation $2x = cosx + 3$ correct to three decimal places by using iteration method.	CLO3	[4]
Q4.	(a)	What is bisection method?	CLO1	[1]
	(b)	Establish the Newton- Raphson method for solving a polynomial equation. Also, show that this has quadratic convergence.	CLO2	[5]
	(c)	Use Newton's method to find a root of the equation $x^3 - 3x - 5 = 0$.	CLO3	[4]
Q5.	(a)	What do you mean by numerical integration?	CLO1	[1]
	(b)	Derive the Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rules for numerical integration.	CLO2	[5]
	(c)	Compute the integral $\int_0^1 \frac{dx}{1+x^2}$ up to 5 decimal places by Simpon's $\frac{1}{3}$ and $\frac{3}{8}$ rules. Compare the results with the correct value and comment on the results (assume $\pi = 3.14159$.)	CLO3	[4]

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Date: 30/06/2025

Semester: Spring 2025
Time: 1 hour 30 minutes
Total Marks: 30

Mid-Term Examination

Answer any **five** questions from the following: 5*6 = 30

1 (a) Describe Computer Peripherals and Computer Interfacing with examples. 3
(b) What is Data Bus? Draw the diagram of Bus in the computer system. 3

2 (a) Suppose you are typing on the keyboard. One key is displaying many times simultaneously when you press the key. What do you think, why is this happening? How can you solve this problem? 2
(b) What is Bounce effect? Draw a flowchart of keyboard processing. 4

3 (a) What is Asynchronous and Synchronous serial data? 2
(b) Suppose that an asynchronous system requires seven bits(10), two stop bits (0 for one, 1 for 1.5 or 2 stop bits), odd parity, and a baud rate 8400. Configure the line control register. 2
(c) Suppose you have divisor value 240, what will be the Baud rate for asynchronous serial data? 2

4 (a) Describe the Hall effect Keyswitch. What is the difference between EPROM and EEPROM . 3
(b) Why seven segment display is known as alpha numeric display? Draw a seven-segment LED display for **25B**. 3

5 (a) Write down the primary components of a hard disk. 2
(b) Find out the base current and voltage drop through the register for the three display **(157)**. 2
(c) Write about Memory-mapped I/O and handshaking. 2

6 (a) Define: 3
i. Read Only Memory(ROM) discs.
ii. Once Read Many (WORM) discs.
iii. Re-writable discs.
(b) What is CRT? Explain Interlaced and non-interlaced principles. 3

7 (a) Outline the instructions of a digital computer. 3
(b) Illustrate the connection between the interface unit and the highway in a diagram 3

2 5 6

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323: Operating System

Batch: 23rd

Semester: Spring 2025

Date: 28/06/2025 **Mid-Term Examination**

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions of the following. **$5 \times 6 = 30$**

1 (a) Explain the role of Operating System (OS) as "**an extended machine**" and "**a resource manager**". 3

(b) Briefly explain about Batch operating system and Multitasking OS 3

2. (a) Explain the process of converting Physical address to logical address with diagram. 3

(b) Determine Physical Address from the following table 3

Process	Limit Register	Relocation Register	Logical Address	Physical Address
P_0	380	700	210	
P_1	425	1100	500	
P_2	115	200	80	

3. (a) Write the criteria of CPU Scheduling. 2

(b) Processes are given with arrival and burst time on the following table. The time quantum of the system is **3 units**. Apply Preemptive Round Robin Scheduling Algorithm and answer the following questions. 4

Process ID	Arrival Time	Burst Time
P1	0	7
P2	1	4
P3	2	15
P4	3	11
P5	4	20
P6	4	9

i. Draw the Gantt chart with Ready Queue.
ii. Calculate average Waiting time.
iii. Calculate average Turn Around time.
iv. Calculate Throughput in percentage.

4. (a) What is process? Explain Process State Transition with figures. 3

(b) What is PCB? Explain different kinds of Threads with diagram? 3

5. (a) What is Kernel? Explain the dual mode operation with figure. 3

(b) Elaborate the following terms with diagram: 3

i. Swapping
ii. Fragmentation
iii. Partitioning

6. (a) What is system calls? When CPU switches to another process the system must save the state of the old process and need to reload it for new execution. Now draw and explain the diagram for this task which will be done by Operating System. 3

(b) From the concept of fork() system call, determine how many times the printf functions will be executed in the following program. Explain your answer using a tree hierarchy. 3

```
#include <stdio.h>
int main(){
    fork();
    fork();
    printf("Child process one");
    fork();
    printf("Child process two");
    return 0;
}
```

7. In the following table, there are 7 processes with their arrival time and burst time. The time quantum of the system is **2 units**. 6

Process ID	Priority	Arrival Time	Burst Time
1	2	0	3
2	6	2	5
3	3	1	4
4	5	4	2
5	7	6	9
6	4	5	4
7	10	7	10

Now calculate waiting time by using both FCFS and Priority Scheduling Algorithms (non-preemptive) and determine which algorithm is better in terms of faster process execution.

higest number
lowest priority

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology
Semester Final Examination 2025

Course Code: MAT323
Final Examination
Date: 9/10/2025

Course Title: Numerical Analysis
Semester: Spring 2025
Total Time: 02:00 Hours
Total Marks: 40

Answer any four (04) out of the following questions: $4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	**(a)**	What is curve fitting?	CLO1	[1]
	(b)	Fit a second degree curve to the following data taking x as the independent variable:	CLO3	[4]

x:	0	1	2	3	4
y:	1	5	10	22	38

	(c)	Fit an exponential curve of the form $y = ab^x$ to the following data:	CLO3	[5]

x:	1	2	3	4	5	6	7	8
y:	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Q2.	**(a)**	Discuss 'Gauss- Seidel' iteration method for solving the system of linear equations.	CLO2	[4]
	(b)	Find the solution of the system $83x + 11y - 4z = 95$, $7x + 52y + 13z = 104$, $3x + 8y + 29z = 71$, using Gauss- Seidel method up to three decimals.	CLO3	[6]
Q3.	**(a)**	Define eigen values and eigen vectors.	CLO1	[2]
	(b)	What is Euler's method?	CLO1	[2]
	(c)	Find the largest eigen values and the corresponding eigen vectors of the following matrix using power method.	CLO3	[6]

$$\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$$

Q4.	**(a)**	What is the initial value problem?	CLO1	[1]
	(b)	Derive fourth order Runge-Kutta method for solving first order ordinary differential equation.	CLO2	[4]
	(c)	Use Runge-Kutta method to approximate y, when $x = 0.1$ and $x = 0.2$, given that $x = 0$ when $y = 1$ and $\frac{dy}{dx} = x + y$.	CLO3	[5]
Q5.	**(a)**	Solve the following system of linear equations by the Gauss Jordan method. $2x + 3y - z = 5$, $4x + 4y - 3z = 3$, $-2x + 3y - 2z = 1$.	CLO3	[4]
	(b)	Solve the following system of linear equations by the Factorization method. $x + 5y + z = 21$, $2x + y + 3z = 20$, $3x + y + 4z = 26$	CLO3	[6]
Q6.	**(a)**	Describe the Picard's method of successive approximations for the solution of a differential equation.	CLO2	[4]
	(b)	Using Picard's method, obtain the solution of $\frac{dy}{dx} = x(1 + x^3y), y(0) = 3$.	CLO3	[3]
	(c)	Solve by Euler's method of the equation $\frac{dy}{dx} = x + y, y(0) = 1$, at the points $x = 0.05$ and $x = 0.10$ taking $h = 0.05$.	CLO3	[3]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Batch: 23rd

Semester: Spring 2025

Date: 20/09/2025 **Final Examination** Time: 2 hours

Total Marks: 40

Answer any **five** questions from the following: 5x8 = 40

1	(a)	Outline the need of PPI.	3
	(b)	Draw the block diagram of 8155 with the help of 8085.	5
2	(a)	Define Memory Mapped I/O.	2
	(b)	Draw the block diagram of 8255 and write about the data bus buffer of 8255	6
3	(a)	Draw the block diagram of Parallel Interface.	3
	(b)	Explain the concept of Small Computer System Interface(SCSI) and draw the possible SCSI system: multiple initiators, multiple targets.	5
4	(a)	Explain the functions of the bus management lines in GPIB (IEEE 488).	4
	(b)	Provide an overview of the uses for 7-segment display and draw the table for displaying 26C in 7-segment display.	4
5	(a)	Describe the advantages of Stepper motor.	4
	(b)	State the differences between Stepper motor and Servo motor.	4
6	(a)	Describe the applications of Stepper motor.	4
	(b)	Explain about MICR, OCR, and OMR.	4
7	(a)	Suppose Vout = 6V, R1 = 3.7 Ω and R2 = 8 Ω. Find out Vin for Non-inverting Amp.	2
	(b)	Draw the flow graph of software keyboard interfacing. Write down the principles of raster scan.	6

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323: Operating System

Batch: 23rd

Semester: Fall 2024

Date: 17/09/2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the following.. **$5 \times 8 = 40$**

1 (a) Define deadlock and explain the four necessary conditions that must be satisfied simultaneously for a deadlock to occur. 4

(b) Consider the data where Processes: P1, P2, P3, Resources: R1, R2 Allocations: P1 → R1 (Allocated), P2 → R2 (Allocated), P1 → R2 (Requested), P2 → R1 (Requested). 4

Determine whether this system is in a deadlock. Justify your answer.

2. (a) Define the following terms related to disk performance: 4

   i. Seek time
   ii. Rotational latency
   iii. Transfer rate
   iv. Average access time.

(b) Explain Moving-head Disk Mechanism with figure 4

3. (a) What is Shell Script? Why do we need Shell Script? 3

(b) Define how the followings are defined in shell script. 5

   i. If-Else
   ii. Case
   iii. For Loop
   iv. Shell Arithmetic
   v. Variables

4. (a) What is a safe state in the context of deadlock avoidance? 2

(b) When a process requests a resource, what condition must the system check before granting the request in the resource allocation graph method of deadlock avoidance? 3

(c) Distinguish between a safe state, an unsafe state, and a deadlock state with the help of a diagram. 3

5. (a) Explain the purpose of the Banker's Algorithm and describe the data structures required for its implementation. 3

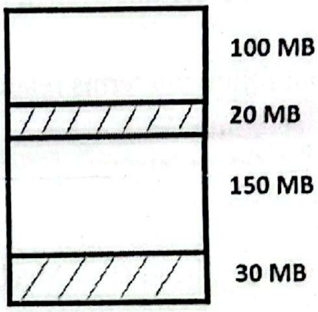
(b) Given the following system state: Determine the safe sequence using Banker's Algorithm 5

	Allocation	Max	Available
	A B C	A B C	A B C
P0:	0 1 0	7 5 3	3 3 2
P1:	2 0 0	3 2 2	
P2:	3 0 2	9 0 2	
P3:	2 1 1	2 2 2	
P4:	0 0 2	4 3 3	

6. (a) What are Contiguous and Non-Contiguous Memory Allocation? 3

(b) Apply First-fit, Next-fit, Best-fit, Worst-fit for the following example and Justify which Algorithms is better for variable size partitioning. 5

Process ID	Size
P1	10 MB/50
P2	15 MB/75
P3	20 MB/100
P4	7 MB/25
~~P5~~	~~8 MB~~



Primary Memory Partition

7. (a) Explain Hardware Architecture of Paging with figure. 3

(b) Solve the followings mathematical problems: 5

i. If Logical Address is 24bits, find out the Logical Address Space.

ii. A process size is 8MB, Physical Address Space is 16MB. If the Page Size is 4KB then calculate Total no of Page and Total no of Frames from the given information.

iii. Virtual Address Space is 4GB and Physical Address is 32bits. If the Page Size is 23 bits then calculate Total no of Page and Total no of Frames from the given information.

1KB = 2^{10} 1MB = 2^{20} 1 GB = 2^{30}

1TB 2^{40}

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 321: Theory of Computing

Batch: 23rd | Semester: Spring 2025

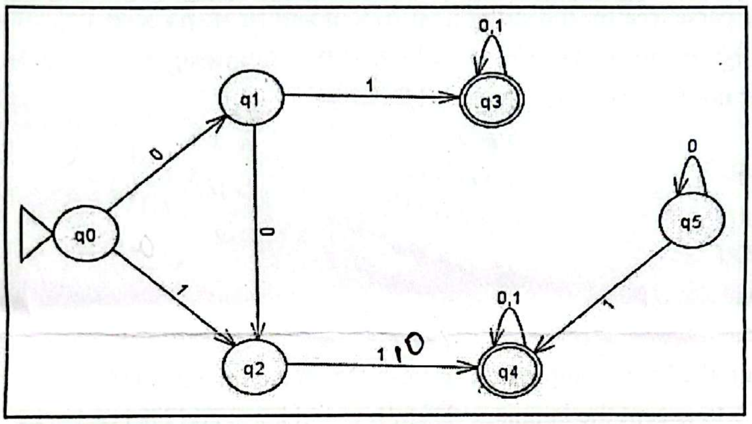
Time: 2 Hours

Date: 08/09/2025 | **Final Examination** | Total Marks: 40

Answer five questions of the following. $5 \times 8 = 40$

1(a). Consider the language **L= {w|w contains at least three 1s}**. Write a regular language for the given formal language L. **2**

1(b). Consider the following Deterministic Finite Automata (DFA): **6**



Convert the given DFA to minimized DFA.

2(a). Define the formal definition of Context-Free Grammar (CFG). **2**

2(b). Answer each part for the following context-free grammar G: $6 \times 1 = 6$

$$R \rightarrow XRX \mid S$$
$$S \rightarrow aTb \mid bTa$$
$$T \rightarrow XTX \mid X \mid \varepsilon$$
$$X \rightarrow a \mid b$$

a. What are the variables of G?
b. What are the terminals of G?
c. Which is the start variable of G?
d. Find three strings in L(G).
e. Find three strings not in L(G).
f. True or False: $T \Rightarrow aba$.

3(a). Design a Moore machine to generate 1's complement of a given binary number. **4**

3(b). Design a Mealy machine that counts the occurrence of the sequence "abb" in any input string over {a,b}. 4

4(a). What are the rules for a Context-Free Grammar to be in Chomsky Normal Form? 2

4(b). Consider the following grammar : 6

$S \rightarrow ASB$
$A \rightarrow aAS \mid a \mid \varepsilon$
$B \rightarrow SbS \mid A \mid bb$

Convert this grammar to an equivalent Chomsky normal form.

5(a). Consider a grammar G: 3

$\langle EXPR \rangle \rightarrow \langle EXPR \rangle + \langle TERM \rangle \mid \langle TERM \rangle$
$\langle TERM \rangle \rightarrow \langle TERM \rangle \times \langle FACTOR \rangle \mid \langle FACTOR \rangle$
$\langle FACTOR \rangle \rightarrow (\langle EXPR \rangle) \mid a$

Construct the parse tree for the string $a + a \times a$ and $(a + a) \times a$.

5(b). Define Ambiguity in CFG. Check whether the following grammar is ambiguous or not for string w = 'aabbccdd':

$S \rightarrow AB / C$
$A \rightarrow aAb / ab$
$B \rightarrow cBd / cd$
$C \rightarrow aCd / aDd$
$D \rightarrow bDc / bc$

S → AB
→ aAbB
→ aabbB
→ aabbcBd
→ aabbccdd

S → C
→ aCd
→ aaDdd
→ aabDcdd
→ aabbccdd

6(a). Briefly explain the basic components of Push Down Automata (PDA). 3

6(b). Design a PDA to accept the language $L=\{a^i b^j c^k \mid i,j,k \geq 0 \text{ and } i = j \text{ or } i = k\}$. 5

7(a). Write the formal definition of Turing Machine. 3

7(b). Design a Turing machine that takes two numbers as input N_1 and N_2 of equal length in binary and computes the logical OR of the two numbers. The tape initially contains $N_1\#N_2\#$, where '#' is a tape symbol that is used as the separator. 5

Operating System Lab (Final Exam)

Time: 50 Minutes
Course Title: Operating System Lab
Course Code: CSE 324

Question 1 (10 marks)

Write a shell script that will take n numbers as input from the user and then:
1. Calculate the sum of all numbers.
2. Calculate the product of all numbers.
3. Find the largest and smallest number among them.
4. Calculate the average of all numbers.

Finally, display all the results in a clear format.
Hint: You may use a for loop to read numbers one by one and perform calculations.

Question 2 (10 marks)

Write a shell script to check whether a given number is:

- Positive
- Negative
- Zero

Print a suitable message for each case.

Question 3 (10 marks)

Write a shell script using a `for` loop to print the **multiplication table** of a number entered by the user.
Example: If user enters 5, print the table from 5×1 up to 5×10.

Question 4 (10 marks)

Menu-driven Calculator

Write a shell script that shows the following menu to the user:

1. Addition 2. Subtraction 3. Multiplication 4. Division 5. Exit

Take two numbers as input.

Ask the user to choose an option (1–5). Use a case statement to perform the chosen operation.

Display the result. Exit when the user chooses option 5.